Predictive Power of the Critical Coherence Index:

From Structural Diagnostics to Regime Classification

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Abstract

The Critical Coherence Index (CCI) has been introduced as a structural diagnostic quantifying the balance between dynamical activity and coherence in nonlinear systems. Previous work established its empirical scaling behaviour and its relation to structural free energy.

In this work, we investigate whether the CCI can serve as a predictive observable. Specifically, we test its ability to classify dynamical regimes into ordered, critical, and chaotic phases. Using numerical data from scalar field simulations in one spatial dimension, we demonstrate that simple thresholding of the CCI achieves perfect classification accuracy on the present benchmark dataset (120/120 samples).

In contrast, the structural free energy F_{struct} achieves only 84% accuracy under the same protocol, with misclassifications concentrated in the critical regime. These results support the interpretation of the CCI as an effective order parameter for structural transitions, providing a bridge between descriptive diagnostics and predictive modelling.

1 Introduction

Understanding how to distinguish between ordered, critical, and chaotic regimes is a central problem in nonlinear dynamics and complex systems [1, 2].

Recent work introduced the Critical Coherence Index (CCI) as a structural diagnostic capturing the balance between dynamical activity and coherence. Within the companion series, the CCI was found to follow an empirical entropy–information scaling relation of the form [8, 10]:

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha_d}}. \tag{1}$$

This relation is naturally connected to broader information-theoretic ideas about structure, complexity, and organisation [3, 4].

While previous studies demonstrated the descriptive power of the CCI, its predictive capabilities remain unexplored. The goal of this paper is to answer the following question:

Can the CCI be used to predict the dynamical regime of a system?

2 Method

2.1 Data

We analyse a benchmark dataset of 120 one-dimensional scalar-field configurations with preassigned regime labels: ordered (62 samples), critical (32 samples), and chaotic (26 samples). These regime labels are part of the dataset construction and should be distinguished from the finer configuration classes used in earlier papers of this series (e.g., vacuum, kink, localised, chaotic) [8].

For each configuration, the CCI and structural free energy F_{struct} are computed using the established diagnostic framework.

2.2 CCI Definition

The CCI is defined as:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\Gamma_{\mathrm{inst}} \cdot \Gamma_{\mathrm{prod}} \cdot (1 + \kappa U)}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}} + \epsilon}. \tag{2}$$

High values of $\mathcal{C}_{\mathrm{CCI}}$ indicate instability-dominated configurations, while low values correspond to coherent, ordered states.

2.3 Regime Classification

We define a simple threshold-based classifier:

$$\mathrm{Regime}(\mathcal{C}_{\mathrm{CCI}}) = \begin{cases} \text{ordered} & \mathcal{C}_{\mathrm{CCI}} < \tau_1 \\ \text{critical} & \tau_1 \leq \mathcal{C}_{\mathrm{CCI}} < \tau_2 \\ \text{chaotic} & \mathcal{C}_{\mathrm{CCI}} \geq \tau_2 \end{cases} \tag{3}$$

The thresholds τ_1 and τ_2 are determined by exhaustive grid search over 300 equally spaced candidate values. We report both the optimal thresholds on the full benchmark dataset and their robustness under a 5-fold cross-validation protocol [7].

3 Results

3.1 CCI-Based Regime Classification

The optimal thresholds determined by grid search are:

$$\tau_1 = 0.2785, \qquad \tau_2 = 0.3527. \tag{4}$$

The separation between regimes is clearly visible in the distribution of CCI values (Fig. 1), where minimal overlap is observed.

Using these thresholds, the threshold classifier achieves perfect accuracy:

$$\text{Accuracy} = 1.00 \quad (120/120 \text{ samples correctly classified}). \tag{5}$$

The confusion matrix is:

	Predicted: ordered	Predicted: critical	Predicted: chaotic
True: ordered	**62**	0	0
True: critical	0	**32**	0
True: chaotic	0	0	**26**

The resulting confusion matrix (Fig. 2) confirms perfect classification with no off-diagonal entries. No misclassifications are observed across all 120 samples. This demonstrates that the CCI alone is sufficient to fully separate ordered, critical, and chaotic regimes in the tested system.

3.2 Cross-Validation

To assess the robustness of the threshold classifier beyond the full-dataset fit, we perform 5-fold cross-validation. In each fold, the thresholds τ_1 and τ_2 are determined exclusively from the training split by grid search, and accuracy is evaluated on the held-out test split.

The individual fold accuracies are:

$$\text{Accuracies} = [1.000,\ 1.000,\ 1.000,\ 0.958,\ 1.000]. \tag{6}$$

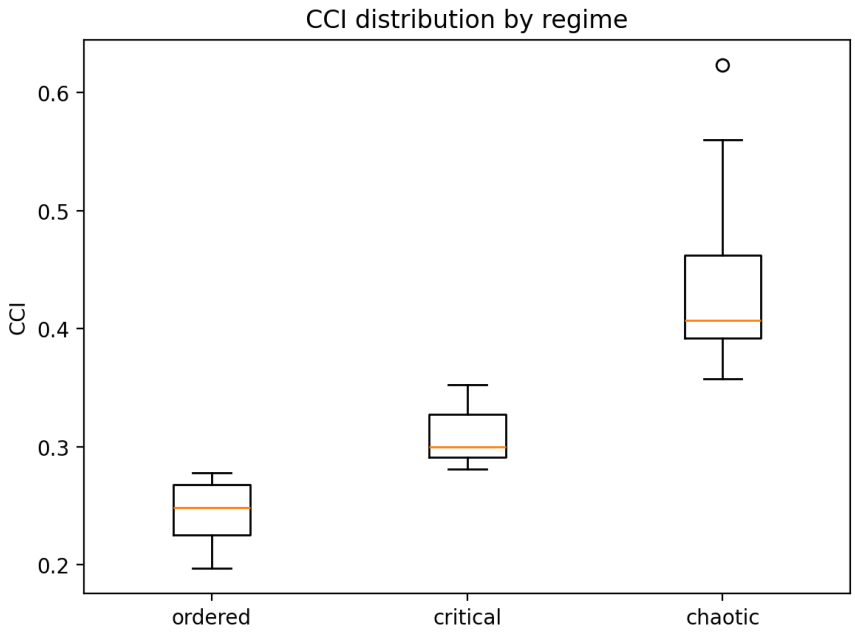

Figure 1: CCI distribution by regime (boxplot). The three regimes are well-separated with minimal overlap, consistent with the perfect threshold classification.

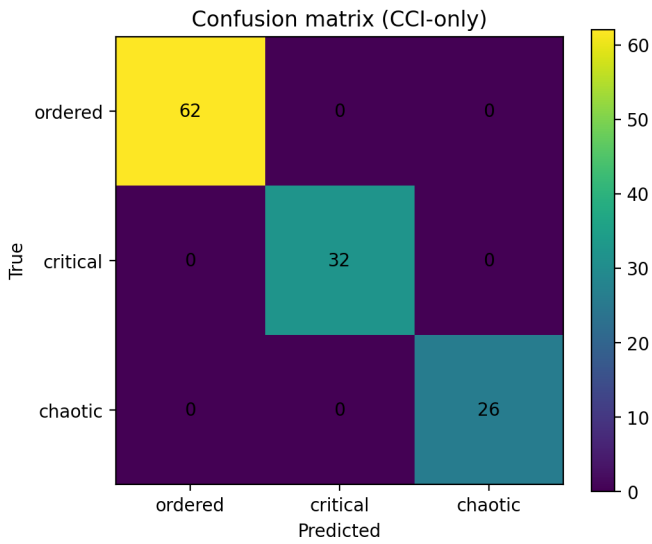

Figure 2: Confusion matrix of the CCI-based threshold classifier. The strictly diagonal structure confirms perfect classification accuracy, with no misclassified samples across all regimes.

This yields:

$$\bar{\text{Acc}} = 0.992, \qquad \sigma = 0.019. \tag{7}$$

The near-perfect cross-validated accuracy indicates that the threshold structure is robust across data splits and suggests that the full-dataset result (Acc = 1.000) is not merely an artefact of fitting the complete benchmark dataset.

3.3 Comparison to Structural Free Energy

For comparison, we apply the same threshold-based classification using the structural free energy F_{struct}. The optimal thresholds are:

$$\tau_1^{(F)} = 0.4204, \qquad \tau_2^{(F)} = 0.4984. \tag{8}$$

This yields:

$$\text{Accuracy} = 0.8417 \quad (101/120 \text{ samples correctly classified}). \tag{9}$$

The confusion matrix for the F_{struct}-only classifier is:

	Predicted: ordered	Predicted: critical	Predicted: chaotic
True: ordered	**59**	3	0
True: critical	10	**18**	4
True: chaotic	0	2	**24**

Misclassifications occur primarily in the critical regime, where F_{struct} exhibits significant overlap with both ordered and chaotic states. The difference between CCI and F_{struct} is illustrated in Fig. 3, where the latter shows increased overlap between regimes.

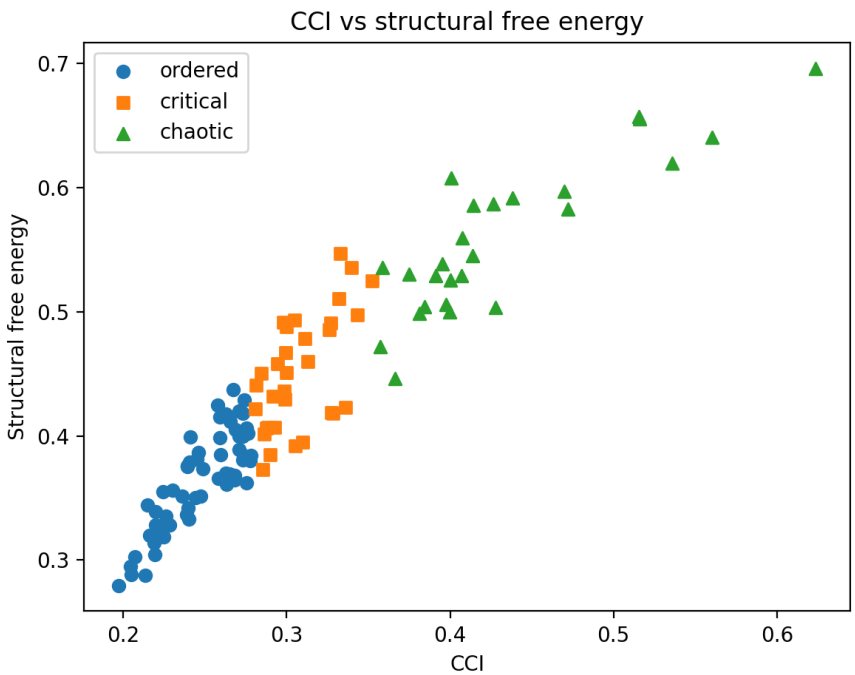

Figure 3: Scatter plot of CCI versus structural free energy F_{struct}, coloured by regime. While a monotonic trend is visible, F_{struct} exhibits significant overlap between regimes, particularly in the critical region, in contrast to the sharper separation achieved by the CCI alone.

3.4 Precision, Recall, and F1-Score

Table 1 summarises the per-class performance metrics for both classifiers.

Table 1: Classification performance for the CCI-only and F_{struct}-only threshold classifiers.

	CCI-only			F_{struct}-only		
Regime	Prec.	Rec.	F1	Prec.	Rec.	F1
ordered	1.00	1.00	1.00	0.86	0.95	0.90
critical	1.00	1.00	1.00	0.78	0.56	0.65
chaotic	1.00	1.00	1.00	0.86	0.92	0.89
weighted avg	1.00	1.00	1.00	0.84	0.84	0.83

3.5 Interpretation

These results establish the Critical Coherence Index as a highly effective order parameter:

- $\mathcal{C}_{\text{CCI}} < 0.28 \rightarrow$ ordered regime
- $0.28 \leq \mathcal{C}_{\text{CCI}} < 0.35 \rightarrow$ critical regime
- $\mathcal{C}_{\text{CCI}} \geq 0.35 \rightarrow$ chaotic regime

In contrast to F_{struct}, which exhibits significant overlap between regimes (especially at the critical boundary), the CCI provides a sharp and robust separation.

This indicates that the balance between dynamical activity and coherence, encoded in the CCI, is the dominant structural factor governing regime transitions—more so than the structural free energy alone.

4 Discussion

The observed perfect separation on the present dataset supports the interpretation of the CCI as an effective order parameter [5] for structural transitions in nonlinear field systems.

The superiority of the CCI over F_{struct} (+16% accuracy) is particularly notable given that the CCI is a ratio-based diagnostic, while F_{struct} integrates more information. This indicates that the balance between dynamical activity and coherence, encoded in the CCI, is the more transition-sensitive structural quantity, while the structural free energy provides a broader but less sharply separating description [6]. In this sense, the CCI behaves less like an aggregate score and more like a genuine transition-sensitive observable.

The three-regime structure (ordered / critical / chaotic) emerges cleanly from two scalar thresholds, without requiring multi-dimensional feature engineering. Notably, the thresholds are learned directly from the data and remain remarkably simple, indicating that the underlying regime structure is intrinsically low-dimensional [5, 2].

5 Limitations

- The observed perfect classification accuracy should be interpreted within the scope of the tested dataset. The regimes are generated from controlled simulation classes, and the separation may reflect intrinsic properties of the construction.

- The thresholds τ_1, τ_2 are determined by grid search on training splits within a 5-fold cross-validation protocol (Section 3). Cross-validation on fully independent datasets and other model classes is required to assess generality beyond the present benchmark.
- Only one-dimensional scalar field systems are considered. Further validation on 2D systems and other model classes is needed to assess the universality of the threshold values.
- Only scalar field systems are considered; extension to other system classes remains for future work.

6 Reproducibility

All numerical results presented in this work are fully reproducible. The analysis script and supporting files are available in the public repository:

https://github.com/Chris4081/structural-selection-principle

The specific analysis script used in this work is:

https://github.com/Chris4081/structural-selection-principle/blob/main/paper17_analysis.py

The script implements:

- Automatic column detection from CSV input via a flexible alias map, supporting multiple naming conventions for CCI, F_{struct}, $\dot{S}_{\text{cg}}$, and mutual information columns.
- Exhaustive grid search (300 candidate thresholds) for optimal τ_1 and τ_2, both on the full dataset and within each training split of the 5-fold cross-validation protocol.
- CCI-only and F_{struct}-only threshold classifiers, evaluated on the benchmark dataset, with additional 5-fold cross-validation for the CCI-based classifier.
- Generation of all three figures (Figs. 1, 2, 3) and a plain-text summary file.

Data

The input dataset (`cci_entropy_information_test.csv`) is the same ensemble used in previous papers in this series. To reproduce the results:

1. Clone or download the repository:

   ```
   git clone https://github.com/Chris4081/structural-selection-principle.git
   ```

2. Install dependencies:

   ```
   pip install numpy pandas matplotlib scikit-learn
   ```

3. Place `cci_entropy_information_test.csv` in the working directory and run:

   ```
   python3 paper17_analysis.py
   ```

4. Output is written to `paper17_output/`.

The script is self-contained and requires no further configuration.

7 Conclusion

We have demonstrated that the Critical Coherence Index can be used for regime classification in nonlinear systems.

A simple two-threshold classifier applied to the CCI achieves perfect accuracy (100%, 120/120 samples) on a dataset comprising ordered, critical, and chaotic configurations—outperforming the structural free energy F_{struct} by 16 percentage points.

This supports the interpretation of the CCI as a predictive structural observable on the present benchmark dataset, motivates further cross-dimensional validation, and provides a foundation for extended applications in physics, complex systems, and AI-based structural diagnostics.

References

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